CAMBRIDGE
International Education

Cambridge O Level

ADDITIONAL MATHEMATICS **4037/21**

Paper 2 **May/June 2024**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

© Cambridge University Press & Assessment 2024

[Turn over

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© Cambridge University Press & Assessment 2024

Mathematics-Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

© Cambridge University Press & Assessment 2024

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method marks, awarded for a valid method applied to the problem.

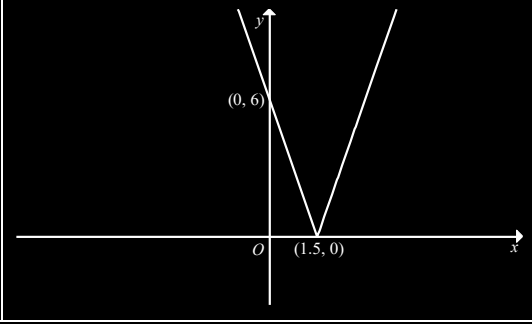
A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.

B Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation 'dep' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt answers which round to
cao correct answer only
dep dependent
FT follow through after error
isw ignore subsequent working
nfww not from wrong working
oe or equivalent
rot rounded or truncated
SC Special Case
soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)	Fully correct graph with intercepts marked 	**B2**	**B1** for a graph of correct shape with vertex on x-axis

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Partial Marks
1(b)	$4x - 6 = 2x$ and $4x - 6 = -2x$ oe	**M1**	
	$x = 3$ $x = 1$	**A2**	**A1** for either correct
	Alternative method		
	$12x^2 - 48x + 36 = 0$ oe	**(B1)**	
	Factorises or solves	**(M1)**	
	$x = 1$, $x = 3$	**(A1)**	
2(a)	$5 - 2(x - 1)^2$	**B3**	Mark final expression **B2** for $-2(x - 1)^2$ or **B1** for $(x - 1)^2$ or $b = -2$, $c = -1$ and **B1** for $5 + b(x + c)^2$ oe with numerical values of b and c or $a = 5$
2(b)	f ≤ *their* 5	**B1**	**STRICT FT** of *their* 5 from **(a)**
3	$5x^2 - 20x + 26$ [= 0]	**M1**	Condone one slip in expansion of brackets or collection of terms
	Correctly finds $b^2 - 4ac$ for *their* $5x^2 - 20x + 26$ [= 0] e.g. $(-20)^2 - 4(5)(26)$	**M1**	**FT** *their* $5x^2 - 20x + 26 = 0$ providing the discriminant is negative for *their* equation
	$400 - 520 < 0$ or -120	**A1**	
	Alternative method		
	$5x^2 - 20x + 26$ [= 0]	**(M1)**	Condone one slip in expansion of brackets or collection of terms
	Completes the square $5(x - 2)^2 + 6$ and states the correct minimum point (2, 6)	**(M1)**	**FT** *their* $5x^2 - 20x + 26 = 0$ providing the minimum point has positive y-coordinate
	Correct conclusion e.g. Minimum point at $y = 6$ therefore does not intersect x-axis oe	**(A1)**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Partial Marks
4(a)	$4(y-3)^2 = 36-9$ or $4y^2-24y+9\,[=0]$	**M1**	
	$y = 3 \pm \sqrt{\frac{27}{4}}$ or exact equivalent, soi	**A1**	
	$3+\sqrt{\frac{27}{4}} - \left(3-\sqrt{\frac{27}{4}}\right)$ oe	**M1**	**FT** *their* $a \pm \sqrt{b}$ providing b is not a square number
	$\sqrt{27}$ or $3\sqrt{3}$ or exact equivalent, nfww	**A1**	
4(b)	Eliminates one unknown and simplifies terms: $2x^2+83 = x^2-20x$ oe, soi	**M1**	
	$x^2+20x+83=0$	**A1**	
	Applies quadratic formula or completes the square: $x = \frac{-20 \pm \sqrt{20^2-4[1](83)}}{2}$	**M1**	**FT** *their* 3-term quadratic
	$x = -10 \pm \sqrt{17}$	**A1**	
	$y = \frac{1}{-10 \pm \sqrt{17}} \times \frac{-10 \mp \sqrt{17}}{-10 \mp \sqrt{17}}$	**M1**	**FT** *their* $x = a \pm \sqrt{b}$ providing previous **M1** awarded
	$x = -10 \pm \sqrt{17}$, $y = -\frac{10}{83} \mp \frac{\sqrt{17}}{83}$	**A1**	**dep** on all marks previously awarded
5(a)(i)	30 240	**2**	**M1**for $3 \times 7! \times 2$ or ${}^3P_1 \times {}^7P_7 \times {}^2P_1$ oe
5(a)(ii)	17 280	**2**	**M1**for $4! \times 6!$ oe or ${}^4P_4 \times {}^6P_5$ oe
5(b)(i)	35	**2**	**M1** for ${}^7C_4$ or 1 + 4 + 18 + 12
5(b)(ii)	51	**2**	**M1** for ${}^3C_2 \times {}^6C_2 + {}^3C_3 \times {}^6C_1$ oe or 18 + 4 + 3 + 2 + 24

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Partial Marks
6	$\frac{\mathrm{d}}{\mathrm{d}x}(\sin^2 x) = 2\sin x\cos x$ soi	**B1**	
	$\cos x \times \textit{their}(2\sin x\cos x) + (\textit{their} - \sin x) \times \sin^2 x$	**M1**	**FT** *their* $\frac{\mathrm{d}}{\mathrm{d}x}(\sin^2 x)$
	$\cos x \times \textit{their}(2\sin x\cos x) + (-\sin x) \times \sin^2 x$ isw	**A1**	**FT** *their* $\frac{\mathrm{d}}{\mathrm{d}x}(\sin^2 x)$
	$\frac{\delta y}{h} = \textit{their}(2\sin 3\cos^2 3 - \sin^3 3)$ or better	**M1**	**FT** *their* derivative
	$0.274h$ or $0.2738[08...]h$ where the coefficient of h is rot to 4 or more sf	**A1**	**dep** on correct derivative seen
7	$\frac{\mathrm{d}y}{\mathrm{d}x} = 2mx + \frac{1}{2}$	**B1**	
	$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = 2m$	**B1**	
	$m = \frac{1}{4}$ and $n = -\frac{5}{4}$ and no other values	**2**	**M1** for $3(2m) = \left(2mx + \frac{1}{2}\right)^2 - \left(mx^2 + \frac{x}{2} + n\right)$ soi

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Partial Marks
8(a)	$S_{30} = \frac{30}{2}\{2a + 29d\} = -1065$	**B1**	
	$S_{50} - S_{30} =$ $\frac{50}{2}\{2a + 49d\} - \left(\frac{30}{2}\{2a + 29d\}\right) = -2210$ or $S_{50} = \frac{50}{2}\{2a + 49d\} = -2210 - 1065$	**B1**	
	Solves *their* linear equations in a and d as far as a = .. or d = … Some correct pairs are e.g. $150a + 2175d = -5325$ $150a + 3675d = -9825$ or $30a + 435d = -1065$ $50a + 1225d = -3275$ or $2a + 49d = -131$ $2a + 29d = -71$	**M1**	**dep** on an attempt to form *their* equations using at least one sum formula
	$a = 8$, $d = -3$	**A2**	**A1** for each
8(b)	$4 + 4r + 4r^2 = 7$ or $\frac{4(1-r^3)}{1-r} = 7$ oe	**B1**	
	$4r^2 + 4r - 3[= 0]$ oe	**B1**	
	Solves or factorises *their* 3-term quadratic oe e.g. $(2r - 1)(2r + 3)$ [= 0]	**M1**	
	$r = 0.5$, -1.5	**A1**	
	$\left[\frac{4}{1-0.5} =\right]$ 8 only, nfww	**A1**	
9(a)	Valid explanation: Range of g is g > 0 oe	**B1**	
9(b)	$\frac{1}{\left(\frac{3x^2}{4x-1}\right)^2}$	**M1**	
	$\frac{(4x-1)^2}{9x^4}$ or simplified equivalent, isw	**A1**	

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Partial Marks
9(c)	$3x^2 - 4xy + y[=0]$ or $3y^2 - 4xy + x[=0]$	**B1**	
	$[x =]\dfrac{-(-4y) \pm \sqrt{(-4y)^2 - 4(3)(y)}}{2(3)}$ oe or $[y =]\dfrac{-(-4x) \pm \sqrt{(-4x)^2 - 4(3)(x)}}{2(3)}$ oe	**M1**	**FT** *their* expression providing it has at most one sign error
	Justifies the negative square root	**B1**	
	$f^{-1}(x) = \dfrac{2x - \sqrt{x(4x-3)}}{3}$	**A1**	
10(a)	$\tan^2 x + 2\tan x \sec x + \sec^2 x$ or $\left(\dfrac{\sin x}{\cos x} + \dfrac{1}{\cos x}\right)^2$	**M1**	
	$\dfrac{\sin^2 x}{\cos^2 x} + 2 \times \dfrac{\sin x}{\cos x} \times \dfrac{1}{\cos x} + \dfrac{1}{\cos^2 x}$ or factorises $\dfrac{1}{\cos^2 x}(\sin x + 1)^2$ oe	**A1**	
	$\dfrac{(1+\sin x)^2}{1-\sin^2 x}$	**A1**	
	$\dfrac{(1+\sin x)(1+\sin x)}{(1-\sin x)(1+\sin x)} = \dfrac{1+\sin x}{1-\sin x}$ or $\dfrac{(1+\sin x)^2}{(1-\sin x)(1+\sin x)} = \dfrac{1+\sin x}{1-\sin x}$	**A1**	must be fully justified
10(b)	$7\sin 3\theta = 5$	**B1**	
	One correct value for 3θ soi e.g. 45.58… 134.4… 405.5… 494.4…	**M1**	
	15.2 or 15.19 to 15.195 44.8 or 44.80 to 44.81 135.2 or 135.19 to 135.195 164.8 or 164.80 to 164.81	**A2**	with no extras in range **A1** for any 2 correct, ignoring extras

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Partial Marks
11	$P = 2\left(\frac{\pi x}{4}\right) + x + 2\left(\frac{400}{x} - \frac{x}{2}\right)$ oe	**B2**	**B1** for rectangle length = $\frac{400}{x}$ soi
	Correct first derivative $\frac{\pi}{2} - \frac{800}{x^2}$	**B1**	**FT** *their P* providing it is of the form $\frac{a}{x} + bx$ oe with *a* an integer and *b* a constant
	Equates *their* $\frac{dP}{dx}$ to 0 and solves for *x*	**M1**	**FT** provided one term of *their* derivative is correct
	$x = \frac{40}{\sqrt{\pi}}$ or 22.6 or 22.56[75...]	**A1**	
	$P = \frac{\pi}{2}\left(\frac{40}{\sqrt{\pi}}\right) + \frac{800}{\frac{40}{\sqrt{\pi}}}$	**M1**	**FT** *their* value of *x* provided it is greater than 0
	P = 70.9 or 70.89[81...]	**A1**	
12(a)	$\overrightarrow{OP} = \frac{4}{7}\mathbf{b} + \lambda\left(\mathbf{c} - \frac{4}{7}\mathbf{b}\right)$ and $\overrightarrow{OP} = \mathbf{b} + \mu\left(\frac{4}{7}\mathbf{c} - \mathbf{b}\right)$	**B3**	**B2** for $\overrightarrow{OP} = \frac{4}{7}\mathbf{b} + \lambda\left(\mathbf{c} - \frac{4}{7}\mathbf{b}\right)$ or $\overrightarrow{OP} = \mathbf{b} + \mu\left(\frac{4}{7}\mathbf{c} - \mathbf{b}\right)$ oe or **B1** for $\overrightarrow{OP} = \left(their\ \frac{4}{7}\right)\mathbf{b} + \lambda\left(\mathbf{c} - \left(their\ \frac{4}{7}\right)\mathbf{b}\right)$ or $\overrightarrow{OP} = \mathbf{b} + \mu\left(\left(their\ \frac{4}{7}\right)\mathbf{c} - \mathbf{b}\right)$ or
	Equates components e.g.: $\left(their\ \frac{4}{7}\right)(1-\lambda) = (1-\mu)$ or $\lambda = their\ \frac{4}{7}\mu$	**M1**	**FT** providing at least **B1** awarded and two expressions for $\overrightarrow{OP}$ in terms of **b**, **c**, λ and μ found
	$\frac{4}{7}(1-\lambda) = (1-\mu) \quad \lambda = \frac{4}{7}\mu$ oe	**A1**	
	$\lambda = \frac{4}{11}$ and $\mu = \frac{7}{11}$ oe and conclusion *AP* : *AC* = 4 : 11 therefore *AP* : *PC* = 4 : 7 *BP* : *BD* = 7 : 11 therefore *DP* : *PB* = 4 : 7 oe	**A2**	**A1** for $\lambda = \frac{4}{11}$ or $\mu = \frac{7}{11}$ oe

© Cambridge University Press & Assessment 2024

Question	Answer	Marks	Partial Marks
12(b)	$\overrightarrow{OP} = \frac{4}{11}(\mathbf{b}+\mathbf{c})$ or $\overrightarrow{OP} = \frac{4}{11}\mathbf{b} + \frac{4}{11}\mathbf{c}$ and $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are scalar multiples of each other and have a point in common oe	**2**	**M1** for $\overrightarrow{OP} = \frac{4}{11}(\mathbf{b}+\mathbf{c})$ or $\overrightarrow{OP} = \frac{4}{11}\mathbf{b} + \frac{4}{11}\mathbf{c}$

© Cambridge University Press & Assessment 2024